PUSHFORWARDS AND HIGHER DIRECT IMAGES OF PARASITIC MODULES ARE PARASITIC

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof displays the presheaf whose sheafification is the higher direct image; the display lies inside the counted span and is reproduced verbatim.

Editorial note. The counted spans carry 2 navigation stubs; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- Higher direct images on a site as the sheafification of the presheaf of cohomology of the base change. Cohomology on Sites, Lemma sites-cohomology-lemma-higher-direct-images, invoked at proof lines 4040-4046. Fixes the presheaf whose vanishing has to be shown.
- Parasitic modules have vanishing cohomology. Same chapter, Lemma lemma-cohomology-parasitic, invoked at proof lines 4048-4049. Gives the vanishing of the displayed cohomology group whenever the base change is flat over T.
- Flatness is preserved by base change. Standard, used at proof lines 4047-4048 and again at proof lines 4050-4051. Makes the base change of a flat morphism flat, both for a single U and for the members of a covering of U.

Source reference. Original lemma 0755: sources/stacks-descent/pinned-original-source.tex, source lines 4005–4005. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 072W: sources/stacks-chow/tags.txt, source lines 9040–9040. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let $f : T \to S$ be a morphism of schemes. For any parasitic $\mathcal{O}$-module on $(Sch/T)_\tau$ the pushforward $f_*\mathcal{F}$ and the higher direct images $R^i f_*\mathcal{F}$ are parasitic $\mathcal{O}$-modules on $(Sch/S)_\tau$.*

Proof. Recall that $R^i f_*\mathcal{F}$ is the sheaf associated to the presheaf

$$U \mapsto H^i((Sch/U \times_S T)_\tau, \mathcal{F})$$

Date: stacks-project:descent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

see Cohomology on Sites, Lemma 072W. If $U \to S$ is flat, then $U \times_S T \to T$ is flat as a base change. Hence the displayed group is zero by Lemma 0755. If $\{U_i \to U\}$ is a τ-covering then $U_i \times_S T \to T$ is also flat. Hence it is clear that the sheafification of the displayed presheaf is zero on schemes U flat over S. $\square$